Thanh Vu

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EDUCATION

Franklin & Marshall College — Lancaster, PA
Bachelor of Arts, Computer Science — *May 2026*

- **GPA:** 3.72/4.0 (Major GPA: 3.82/4.0) | Honors List: Fall 2021, 2023, 2024, 2025 | Dean's List: Spring 2022
- **Relevant Coursework:** Data Science, Algorithms, Artificial Intelligence, Machine Learning, Statistical Modeling, Probability & Statistics, Information Theory, Theory of Computation, Applied Regression Analysis

CERTIFICATIONS

- **Microsoft Certified: Fabric Data Engineer Associate (DP-700)**, Microsoft, Oct 2026
- **SQL (Advanced) Certificate**, HackerRank, Sep 2025

EXPERIENCE

EDC Lancaster County, Center for Regional Analysis — Lancaster, PA
Data Analyst Fellow — *Aug 2026 – Present*

- Built an automated Power BI reporting pipeline that sources labor-market and economic data from federal public APIs (BLS, BEA) into a star-schema model with shared fact and dimension tables, and configured scheduled refresh to replace manual data pulls.
- Developed a Google Apps Script web dashboard backed by Google Sheets as a lightweight data store, with automated cross-sheet imports, nightly refresh, sheet locking, and an integrated Gemini AI assistant that summarizes trends and flags anomalies, so non-technical staff can update it without code.
- Designed data-intake structures and dashboards for partner organizations and internal stakeholders, wrote dataset documentation and data dictionaries, and iterated on requirements with the Director of the Center for Regional Analysis.

KCC Capital Partners — Sheridan, WY
Data Science Intern — *Jan 2026 – May 2026*

- Collected, cleaned, and integrated datasets on precious-metal and mining companies, markets, and transactions; enriched contact records with Seamless.AI to improve data quality for deal sourcing.
- Built statistical and machine-learning models to identify likely strategic and financial buyers for active mandates, and automated recurring reports and dashboards for the deal team.

KPIM Joint Stock Company — Hanoi, Vietnam
Data Engineer Intern — *Jan 2021 – Apr 2021*

- Engineered ETL pipelines in Python (Pandas, NumPy) and SQL Server to automate legacy client data workflows, reducing processing time by 30% and eliminating manual entry errors.
- Validated Power BI reports with the analysis team and documented pipeline logic as reusable technical references.

PROJECTS

Can Mail Data Predict Housing Vacancy? | *Microsoft Fabric (Lakehouse, Dataflow Gen2), Power BI, DAX* — Sep 2026

- Built an end-to-end Fabric pipeline: landed raw USPS quarterly and Census ACS files in a Lakehouse, consolidated them with Dataflow Gen2 (Power Query M) into a county-year table (6 counties, 2010–2024), and served it through a semantic model to Power BI.
- Fixed two silent data-quality bugs (inconsistent source headers; a file-combine step that dropped the county key) and wrote DAX measures, including a hand-built Pearson correlation, showing a pooled r = 0.77 but only 57.7% directional agreement.

Grocery Transaction Data Warehouse | *SQL Server, T-SQL, Medallion Architecture* — Jul 2025 – Aug 2025

- Designed a Bronze → Silver → Gold warehouse over 131M+ rows (1M+ customers, 25K products) with stored-procedure ETL, BULK INSERT loading, load-time logging, and TRY/CATCH error handling.
- Modeled a Gold-layer star schema (fact_sales, dim_customers, dim_products), wrote Silver and Gold data-quality test suites, and documented a data catalog and naming conventions; query tuning cut processing time by 25%.

Sales Planning Analysis | *Power BI, DAX, SQL Server* — Aug 2025

- Integrated sales and monthly plan data from SQL Server into a galaxy-schema model (actual and plan fact tables on shared store and date dimensions) over 58K+ transactions, with DAX measures for plan completion, YoY growth, and rolling 12-month trends.

TECHNICAL SKILLS

Languages: SQL (T-SQL, Spark SQL), Python (Pandas, NumPy, PySpark), KQL, DAX, Power Query M, R, JavaScript (Google Apps Script)
Microsoft Fabric (DP-700): Lakehouse, Warehouse, OneLake, Delta Lake, Data Factory pipelines, Dataflow Gen2, Notebooks (PySpark, Spark SQL), Eventstream, Eventhouse/KQL, semantic models, deployment pipelines
Data Engineering: ETL/ELT pipelines, Medallion architecture, dimensional modeling (star and galaxy schemas), stored procedures, data-quality testing, data documentation
Platforms & Tools: SQL Server, MySQL, Power BI, Power Automate, Google Sheets/Apps Script, Excel, Git/GitHub

LEADERSHIP

Diplomatic Congress, Technology Committee, Franklin & Marshall College — Aug 2021 – Aug 2022

- Led IT policy work aligned with the College's strategic plan, reducing classroom technology costs by 18%, and centralized software license management across 15+ departments.